Construction Chemicals

SUBMITTAL DOCUMENT

O-Guard FR

Water-Based Intumescent
Fire-Protective Coating for Steel



Fire Rating
Up to 120 min

Water-Based
Low odour

Finish
Matt white

Application
Brush, roller, spray

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

O-Guard FR — Water-Based Intumescent Fire-Protective Coating for Steel

Project Name		**Document No.**	OMX-MS-OGUARDFR
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	O-Guard FR	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of O-Guard FR on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers surface preparation, priming check, application and inspection of O-Guard FR intumescent coating on interior structural steel to the thickness required by the fire-protection schedule.

- Inspection and acceptance of the substrate and the work area.
- Surface preparation and repair of defects.
- Mixing and application of the material as described in this document.
- Curing, protection and handover of the completed area.
- Quality control, inspection and documentation of each stage.

3. Definitions

Term	Meaning
Intumescent	Coating that expands in fire to form an insulating char.
Section factor	Ratio of heated perimeter to cross-section area of a steel member.
DFT	Dry film thickness required per member for the fire rating.
Fire rating	Time (minutes) the protected steel keeps its load-bearing capacity.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.

Role	Responsibility
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
O-Guard FR	Intumescent coating, 5 and 20 kg pails
Primer	Approved anti-corrosive primer compatible with O-Guard FR
Top coat	Compatible decorative top coat where specified

5.1 Key product data

Property	Typical Value
Base	Water-based acrylic intumescent
Finish / colour	Matt white
Density	1.3 – 1.4 kg/L
Solids content	65 – 75 %
Touch dry	1 – 2 hours
Recoat interval	6 – 8 hours
Fire rating	up to 120 minutes (interior steel)
Shelf life	12 months

6. Tools & Equipment

- Airless spray pump
- Brushes
- Rollers
- Slow-speed mixer
- Wet-film comb
- DFT gauge
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Measuring buckets and scale
- Masking tape and protective sheeting

- Lighting and extension cables

6.1 Personal protective equipment

- Gloves
- Goggles
- Overalls
- Mask

7. Storage & Handling

- 12 months in original, unopened packaging. Store at +5 °C to +35 °C; protect from frost.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep containers tightly closed and protect from frost and direct sunlight.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Confirm the fire-protection schedule (rating and DFT per member).
2. Check that steel is blast-cleaned and primed with an approved, compatible primer.
3. Clean the primed surface of dust, grease and salts; it must be dry.
4. Pre-wet absorbent surfaces to a saturated surface-dry condition; remove standing water.
5. Protect adjacent surfaces with masking tape and sheeting.
6. Obtain QA/QC approval of the prepared surface before starting the next step.

10. Mixing

1. Check the batch number and expiry date and record them.
2. Stir thoroughly with a slow-speed mixer until uniform.
3. Use as supplied; thin with a little water only if required for spray, as recommended.

11. Application

1. Apply by airless spray, brush or roller in several coats.
2. Check wet film thickness during each coat.
3. Allow 6 – 8 hours between coats.
4. Build up to the DFT required for each member.
5. Work in manageable areas and finish each area in one operation where possible.
6. Check layer thickness and consumption regularly and record the results.

12. Curing & Protection

- Touch dry 1 – 2 hours; recoat after 6 – 8 hours.
- Protect from water and condensation during drying.
- Apply the decorative top coat after full drying where specified.
- Protect from sun, wind and rain; keep moist where required during the first 24 hours.
- Barricade the area and display warning signs.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Primer	Before coating	Approved, cured, clean	QA/QC
Climate	During work	+10 °C to +35 °C, RH ≤ 80 %	Site Engineer
WFT	Each coat	As per coat plan	Applicator
DFT	After drying	≥ required DFT per member	QA/QC + Consultant

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, expiry, packaging condition	Each delivery	Material inspection report
Climate	Air / substrate temperature, RH, dew point	Start of shift and every 4 hours	Daily climate log
Surface preparation	Visual, moisture, soundness	Each area	Inspection request (hold point)
Application	Consumption, thickness, appearance	Each coat / area	Daily progress report
Final inspection	Visual, continuity, finish	Each completed area	Handover checklist

14. Defects & Remedies

Defect	Likely Cause	Remedy

Defect	Likely Cause	Remedy
Shrinkage cracks	Rapid drying; layer too thick; excess water	Remove, reapply in thinner layers and protect from drying
Poor adhesion / hollow areas	Dusty or dry substrate	Remove, clean and pre-wet; reapply
Dusting surface	Excess water or late finishing	Light sanding and apply a fresh thin layer
Uneven finish	Insufficient trowelling	Sand and apply a finishing layer

15. Health, Safety & Environment

Hazard	Control Measures
Skin and eye contact	Gloves, goggles, overalls; eye-wash available
Inhalation of vapours / dust	Ventilation; masks as required
Slips and trips	Housekeeping; barricades and signs
Manual handling	Correct lifting technique; mechanical aids
Electrical tools	Inspected tools; protected cables

- Wear gloves, goggles and a mask during spray application.
- Avoid contact with eyes and prolonged skin contact.
- Ensure ventilation in enclosed areas.
- Eyes: rinse with clean water for 15 minutes and seek medical advice.
- Skin: wash with soap and water.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean tools and spray equipment with water immediately after use.

17. Limitations

- Ambient and substrate temperature: +10 °C to +35 °C.
- Relative humidity: max. 80 %.
- Do not apply in rain or when rain is expected before the material has set.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Fire schedule available	☐ Yes ☐ No	
Primer approved and clean	☐ Yes ☐ No	
Material stirred	☐ Yes ☐ No	
Coats applied	☐ Yes ☐ No	
WFT checked	☐ Yes ☐ No	
DFT measured per member	☐ Yes ☐ No	
Top coat applied (if specified)	☐ Yes ☐ No	
DFT records signed	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

O-Guard FR — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Protective Coating

O-Guard FR

Water-Based Intumescent
Fire-Protective Coating for Steel



Product Description

O-Guard FR is a water-based acrylic intumescent coating for the fire protection of interior structural steel. When exposed to fire it expands to form an insulating char that delays the temperature rise of the steel, providing fire ratings of up to 120 minutes depending on the applied thickness and the steel section.

Uses

- Interior structural steel beams and columns.
- Warehouses and industrial buildings.
- Shopping malls, offices and public buildings.
- Steel structures requiring 60, 90 or 120 minutes fire rating.

Advantages

- Fire protection up to 120 minutes.
- Water-based, low odour.
- Thin-film, decorative white finish.
- Applied by brush, roller or airless spray.
- Can receive a decorative top coat.

Application

1. **Surface Preparation**
   Blast-clean steel and apply an approved anti-corrosive primer; clean and dry before coating.
2. **Mixing**
   Stir thoroughly with a slow-speed mixer; use as supplied.
3. **Application**
   Apply in several coats to the dry film thickness required for the fire rating; apply an optional top coat.

Consumption

1.2 – 1.6 m²/kg
for 60 minutes fire rating
Thickness depends on steel section factor and required rating.

Packaging

5 kg and 20 kg pails.

Storage & Shelf Life

12 months in original, unopened packaging.
Store at +5 °C to +35 °C; protect from frost.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Base	Water-based acrylic intumescent	—
Finish / colour	Matt white	—
Density	1.3 – 1.4 kg/L	—
Solids content	65 – 75 %	—
Touch dry	1 – 2 hours	—
Recoat interval	6 – 8 hours	—
Fire rating	up to 120 minutes (interior steel)	—
Shelf life	12 months	—

Consumption

Fire rating	Coverage
60 minutes	1.2 – 1.6 m²/kg
90 minutes	0.9 – 1.1 m²/kg
120 minutes	0.6 – 0.8 m²/kg

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Ambient and substrate temperature: +10 °C to +35 °C.
- Relative humidity: max. 80 %.
- Interior use only; protect from water and condensation.
- Primer must be compatible and fully cured.

Application Notes

- Required thickness per member must be taken from the fire-protection schedule for the steel section factor.
- Apply in several coats; do not exceed the maximum wet thickness per coat.
- Measure the dry film thickness after each coat.
- A compatible decorative top coat may be applied after full drying.

Health & Safety

- Wear gloves, goggles and a mask during spray application.
- Avoid contact with eyes and prolonged skin contact.
- Ensure ventilation in enclosed areas.

First Aid & Cleaning

- Eyes: rinse with clean water for 15 minutes and seek medical advice.
- Skin: wash with soap and water.
- Inhalation: move to fresh air.
- Ingestion: rinse mouth and seek medical advice.
- Clean tools and spray equipment with water immediately after use.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

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Test Reports

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To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

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To be attached: approvals and certificates, where available.